

Helix: A Decentralized Blockchain-Powered Version Control and Collaborative Storage System

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Introduction

Problem Statement

Centralized providers (Dropbox, Google Drive, OneDrive) control user data, exposing it to outages, censorship, breaches, vendor lock-in, and privacy erosion through scanning or compliance demands. Traditional version control like Git excels locally but depends on centralized platforms for collaboration and remote storage. In an era of data sovereignty, users need immutable, transparent, and resilient alternatives for collaboration, especially in academia, journalism, and sensitive workflows.

Project Introduction

This paper presents Helix, an end-to-end decentralized storage and linear version control system. It stores files on IPFS (content-addressed, immutable) and records ownership and version metadata on a blockchain via smart contracts. Helix offers a user-friendly, responsive web interface for seamless file uploads, linear version tracking, and retrieval. By leveraging modern Web3 tooling, it provides a production-ready, high-performance demonstration of blockchain and IPFS integration for individuals and small teams.

Objectives

- Design and implement a hybrid architecture that stores file content on IPFS (immutable, content-addressed) while recording ownership and linear version history on a blockchain via smart contracts.
- Develop a responsive, modern web DApp using Vite and React, integrated with MetaMask for secure wallet-based authentication and transaction signing.
- Implement a simplified, linear version history (Version 1 → Version 2 → Version 3) to track file evolution immutably on-chain, providing a clear audit trail.
- Utilize Foundry for blazing-fast smart contract development, testing, and local blockchain simulation, ensuring a robust and optimized backend logic.
- Deliver a production-ready demonstration on an Ethereum-compatible testnet that proves the feasibility of a streamlined blockchain + IPFS collaboration tool.
- Evaluate technical, operational, and economic feasibility, documenting limitations (gas costs, pinning persistence) for future improvements.

Scope

The project focuses on a functional web-based decentralized file system targeting individual and small-team collaboration. It covers local file ingestion to IPFS, on-chain metadata management, linear versioning, and a responsive UI.

In-scope:

- Responsive Web DApp with MetaMask integration for file upload, version tracking, and retrieval.
- IPFS storage integration via REST API (e.g., Pinata) for reliable file pinning and CID retrieval.
- Smart contract logic for ownership registration and immutable linear version linking.
- Linear version history viewer and basic user dashboard.
- Testnet deployment and responsive web interface.

Out-of-scope:

- Client-side AES encryption (files are uploaded in plaintext for V1 to simplify core architecture and focus on the storage/versioning pipeline).
- Complex Git semantics (branching, merging, conflict resolution).
- Fine-grained, multi-user access control (V1 focuses on single-owner linear history).
- Full peer-to-peer node operation or incentive layers (e.g., Filecoin).
- Mobile-native applications.

Literature Review

1. Gowda et al. (2025) demonstrated blockchain–IPFS integration for secure decentralized file sharing, showing improved integrity, availability, and resistance to centralized failures—directly supporting Helix’s core architecture.
2. Haque et al. (2025) proposed an integrated blockchain–IPFS solution for secure source-code repository hosting with version control, hybrid cryptography, and smart-contract access control, informing Helix’s Git-like versioning and permission model.
3. Tianqi et al. (2026) examined on-chain/off-chain collaborative storage using blockchain and IPFS, analyzing efficiency trade-offs in metadata management and retrieval that are relevant to Helix’s hybrid design.
4. Han et al. (2025) introduced the IPFS-BC system for collaborative storage, demonstrating practical multi-user access control and IPFS–blockchain synergy aligned with Helix’s small-team goals.
5. Collective authors (2025) evaluated smart-contract-based access control for IPFS file sharing, validating client-side encryption and on-chain permission patterns used in Helix.
6. Various authors (2025) studied decentralized document and package version control on Ethereum + IPFS, providing benchmarks for provenance, tracking, and revert mechanisms.
7. Web3 research surveys (2025–2026) reviewed production patterns for pinning, metadata handling, and user-centric interfaces in decentralized storage DApps, guiding Helix’s UI/UX and deployment choices.
8. Protocol Labs & contributors (2025) detailed advancements in IPFS content addressing, interoperability, and censorship resistance that underpin Helix’s storage layer.
9. GeeksforGeeks contributors (2025) described practical techniques for sharing files via IPFS and blockchain, including version tracking and encryption.

10. Coinbase Learn (2025) provided an overview of IPFS–blockchain complementarity, persistence models, and dApp use cases relevant to decentralized storage platforms.
11. Benet (2014) introduced the InterPlanetary File System, establishing the content-addressed, distributed foundation that Helix builds upon.
12. Nizamuddin et al. (2019) presented a blockchain-based framework for secure IPFS file sharing and access control, an early demonstration of the hybrid model.
13. Kumar et al. (2020) explored blockchain-enabled decentralized storage with IPFS for data integrity and availability.
14. Naz et al. (2021) proposed a secure data-sharing scheme combining IPFS and Ethereum smart contracts with attribute-based access control.
15. Vimal & Srivatsa (2022) designed a blockchain–IPFS hybrid system focused on privacy-preserving collaborative document management.
16. Sharma et al. (2023) analyzed performance and gas-cost implications of storing version metadata on-chain versus off-chain while keeping content on IPFS.
17. Li et al. (2023) introduced a decentralized version-control system using Ethereum and IPFS that supports branching and merging semantics.
18. Chen et al. (2024) evaluated client-side encryption strategies and key-management approaches for IPFS-based collaborative platforms.
19. Patel & Mehta (2024) surveyed smart-contract patterns for fine-grained permission management in decentralized storage applications.
20. Zhang et al. (2025) examined resilience and availability guarantees of IPFS pinning services when combined with blockchain audit trails, highlighting practical deployment considerations.

Proposed Methodology

Development

1. Research & Requirements – Literature analysis, stakeholder needs, and system design.
2. Architecture Design – Definition of storage, blockchain, and presentation layers; smart-contract interfaces.
3. Smart-Contract Development – Solidity contracts for ownership and linear version linking; compiled, tested, and deployed locally using Foundry.
4. IPFS Integration – File upload to IPFS via REST API (Pinata or local node), retrieving the Content Identifier (CID) and ensuring persistence.
5. Frontend & Wallet Integration – Development of a responsive web DApp using Vite + React. Integration of MetaMask via wagmi and RainbowKit for seamless wallet connection, transaction signing, and balance tracking.
6. Testing & Security Review – Unit and integration tests for smart contracts (Foundry).
7. Testnet Deployment & Evaluation – Deployment on an Ethereum-compatible testnet, usability testing, and performance measurement.

Core Workflow

Select File → Upload to IPFS via REST API → Receive CID → Trigger MetaMask Transaction via wagmi → Smart Contract appends CID to linear history array → Frontend UI updates with new version.

System Architecture

- Storage Layer: Files are stored on IPFS. Content Identifiers (CIDs) guarantee immutability and content addressing. Pinning services ensure persistence.

- **Blockchain Layer:** Solidity smart contracts on an Ethereum-compatible testnet maintain ownership and linear version chains (CID linkages). Managed and deployed via Foundry.
- **Application Layer:** A responsive React web application (bundled via Vite) utilizing wagmi/viem for blockchain interaction and TailwindCSS for a modern UI.
- **Interaction Flow:** User authenticates via MetaMask → selects and uploads file → CID recorded on-chain with ownership → subsequent versions create new CIDs linked to previous ones → authorized users retrieve and view history.

Modules Description

- **Module 1: User Authentication & Wallet Integration** – Connects to MetaMask using RainbowKit/wagmi. Derives user identity from the connected wallet address, manages session state, and handles network switching.
- **Module 2: IPFS File Ingestion** – Handles local file reading and uploads the file to an IPFS pinning service via HTTP REST API, returning the immutable CID.
- **Module 3: Smart-Contract Interaction** – Utilizes wagmi React hooks to interact with the Foundry-deployed Solidity contracts. Handles ownership registration, version linking, and event emission directly from the frontend.
- **Module 4: Linear Version Control & History** – Fetches the array of CIDs from the blockchain and displays them as a chronological, immutable timeline of file revisions.
- **Module 5: Dashboard & Retrieval** – File listing, version history viewer, and direct retrieval/download of files from IPFS using the stored CIDs.

Feasibility Study

- **Technical Feasibility:** All core components (IPFS REST APIs, Solidity/Foundry, Vite/React, wagmi) are mature, open-source, and highly optimized. Foundry provides superior compilation speed and testing capabilities over legacy tools.
- **Operational Feasibility:** Wallet-based authentication via MetaMask is the industry standard and familiar to Web3 users. The responsive Vite/React web interface requires no special hardware and loads instantly.
- **Economic Feasibility:** Development uses free/open-source tools. Testnet gas costs are negligible. The lightweight nature of Vite and Foundry ensures low operational overhead, making the architecture highly viable for individual/small-team use cases.

Expected Outcome

A fully functional, responsive web DApp that allows users to:

- Seamlessly upload files from their browser to IPFS.
- Maintain immutable, linear version histories tracked on the blockchain.
- View a chronological timeline of file revisions (CIDs).
- Retrieve and download previous versions of files directly from IPFS.

The system will be deployed on a public testnet, accompanied by documentation proving the architectural viability of a modern, streamlined decentralized file manager.

Tools and Technologies Used

Category	Technologies / Tools
Decentralized Storage	IPFS (via Pinata REST API or Local Node)
Blockchain	Solidity, Foundry (forge for contracts, anvil as local blockchain)
Frontend	React (bundled with Vite), TailwindCSS
Rust Libraries	wagmi, viem, RainbowKit (for MetaMask connection)
Wallet / Auth	MetaMask (Browser Extension)
Development & Version	Git, GitHub
Testing	Foundry (Solidity tests), Vitest (Frontend tests)

Timeline

Tasks	Resp	Status	Start	Due	2026			
					Jul	Aug	Sep	Oct
Literature Review & Requirements		Done	7/15	7/31				
Architecture Design & Planning		Doing	7/20	8/8				
Smart Contract Development & Testing		Todo	8/1	9/5				
IPFS Integration + Client-side Encryption			8/2	9/8				
Frontend Development (Next.js + Wallet)			9/1	10/6				
Versioning, Sharing & Dashboard Modules			9/15	10/13				
Integration, Testing & Optimization			10/1	10/15				
Testnet Deployment & Evaluation			10/5	10/15				
Documentation, Report & Finalization			10/10	10/15				

Conclusion

Helix demonstrates that blockchain and IPFS can be combined into a practical, user-friendly system that restores data ownership and provides immutable version control without reliance on centralized providers. By utilizing a modern, high-performance tech stack—specifically Foundry for smart contracts and Vite/React with wagmi for the frontend—Helix achieves faster development cycles, superior testing capabilities, and a seamless user experience. By focusing on a streamlined, linear versioning model for its initial release, the project delivers a robust, production-ready demonstration of decentralized file management, offering a concrete step toward resilient and censorship-resistant collaborative tools.

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